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A comprehensive review of developments and challenges in the 6G internet of things

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Abstract
In order to provide seamless, intelligent, and energy-efficient connectivity, the upcoming sixth generation (6G) of wireless communication seeks to combine edge intelligence, terahertz links, and artificial intelligence. In contrast to its predecessors, 6G is expected to be a network built especially to support vast Internet-of-things (IoT) ecosystems, linking billions of devices in the satellite, aerial, and terrestrial domains. This article offers a thorough analysis of 6G-IoT technologies, covering open research challenges, application scenarios, and enabling architectures. Reviewing how mobile networks have developed toward 6G, it focuses on key enabling technologies like blockchain, artificial intelligence, and reconfigurable intelligent surfaces and looks at how they can help IoT applications like remote sensing, healthcare, industrial automation, and automotive systems. Along with outlining future research directions for the realization of truly intelligent 6G-IoT networks, the paper identifies important challenges pertaining to energy efficiency, security, interoperability, and standardization.

Keywords Terahertz frequencies, AI integration, Quantum communication, Holographic communication, Massive MIMO, Edge computing, Sustainability, Global standardization

1 Introduction

By connecting smart devices, sensors, and machinery to support applications ranging from industrial automation to healthcare and transportation, the Internet of Things (IoT) has emerged as a major enabler of the digital transformation [1–3]. Communication networks are under tremendous strain due to the IoT services' unprecedented demand for connectivity. As the number of connected devices continues to grow exponentially, next-generation communication systems must offer extremely high reliability, ultra-low latency, and global availability while maintaining energy efficiency and security. According to recent research, 6G is going to serve as a convergence network between ubiquitous IoT connectivity, incorporating intelligence, sensing, and sustainability into the network [129, 131].

Features like massive machine-type communication (mMTC), ultra-reliable low-latency communication (URLLC), and enhanced mobile broadband (eMBB) have been

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made possible by fifth-generation (5G) networks [4–6]. However, 5G's design goals will be surpassed by the rapidly expanding number of connected devices and new use cases, like holographic communication, autonomous systems, and the tactile Internet [7–9]. The research community is now looking beyond 5G to the sixth generation (6G) of wireless networks as a result of these challenges.

The forthcoming 6G-IoT paradigm is anticipated to combine sensing, computation, and communication into a single, intelligent ecosystem. It will make use of technologies like edge and fog computing, reconfigurable intelligent surfaces (RIS), terahertz (THz) communication, artificial intelligence (AI), and integrated terrestrial–aerial–satellite connectivity [10–14]. These enablers will allow billions of diverse IoT devices to receive context-aware, autonomous, and energy-efficient services.

In contrast to earlier generations, 6G promises to provide ubiquitous intelligence, semantic communication, and sustainable networking in addition to higher peak data rates [15, 16]. By embedding AI at every layer, 6G-IoT can dynamically optimize resource allocation, adapt to environmental conditions, and guarantee trust and data integrity through distributed security frameworks such as blockchain [17, 18]. 6G networks will therefore move from a connection-centric to an intelligence-centric design, enabling continuous learning and self-organization.

Despite significant progress in 5G and early 6G research, several open challenges remain, including energy efficiency, privacy protection, interoperability among heterogeneous IoT systems, and standardization of space–air–ground–sea integration [19–21]. Addressing these issues is crucial to fully harness the potential of 6G-IoT networks.

The major contributions of this paper are summarized as follows:

- A thorough analysis of how mobile networks evolved from 1G to 6G, emphasizing IoT integration.
- A classification of enabling technologies—such as blockchain, AI, THz, and RIS—and their functions in 6G-IoT.
- A detailed examination of IoT-driven use cases, including UAV-assisted IoT, vehicular IoT, industrial IoT, and healthcare IoT.
- Identification of key research challenges and future research directions for building sustainable and intelligent 6G-IoT systems.

The rest of this paper is structured as follows: Section II presents the evolution from 1G to 6G. Section III outlines the vision of 6G for IoT networks. Section IV discusses the key enabling technologies for 6G-IoT. Section V explores IoT-driven applications and use cases, while Section VI highlights the open research challenges and future directions. Finally, Section VII concludes the paper.

2 Evolution from 1G to 6G

From the early analog era to the upcoming sixth generation (6G), mobile communication systems have evolved in a constant quest for increased data rates, reduced latency, increased dependability, and worldwide connectivity. Communication networks have gradually changed from voice-only systems to intelligent, AI-driven, IoT-enabled ecosystems thanks to innovations introduced by each generation.

In the 1980 s, the first generation (1G) of analog cellular systems—which could only transmit voice—was introduced. Higher spectral efficiency was made possible by digital

modulation and text messaging introduced by the second generation (2G). Fourth-generation (4G) networks brought broadband mobile data and all-IP architecture, while third-generation (3G) systems extended to multimedia and basic Internet services. To support new IoT services, the fifth generation (5G) currently provides massive machine-type communication (mMTC), ultra-reliable low-latency communication (URLLC), and enhanced mobile broadband (eMBB). However, the exponential growth of connected devices, intelligent automation, and data-driven applications calls for a new paradigm that transcends 5G's limitations — the 6G-IoT era.

The major turning points and technological advancements from 1G to 6G are outlined in Table 1, which also shows how each generation has gradually moved closer to massive, intelligent IoT connectivity.

Table 1 presents the summary of quantitative advancements in mobile communication generation on the basis of 3GPP TR 38.913 and ITU-R M.2410 standards. It separates between theoretical peak rates and common rates experienced by the user to avoid overstatement. The other parameters, latency, reliability, device density, and energy efficiency, have to be a priority on the design of IoT, as 6G is required to communicate billions of low-power devices at latitudes below 1 m and at very low probability of outages. The listed metrics are consistent with standardized targets in 3GPP TR 38.913 and ITU-R M.2410 is shown in Fig. 1 [133, 134].

By combining satellite, marine, and aviation components into a single 3D network, the next phase—6G-IoT—will increase connectivity beyond geographical limits. In order to facilitate autonomous network management, predictive optimization, and real-time adaptation for extensive IoT deployments, it will integrate AI-driven intelligence throughout all layers [22–25] (Fig. 2).

Table 1 Evolution of cellular networks toward 6G-IoT

Generation	Approx. launch	Peak data rate*	Typical user rate	Latency	Reliability (BLER)	Device density (devices/km^2)	Energy efficiency	IoT relevance
1G	1980–1990	2.4 kbps (analog)	~2 kbps	≈500 ms	N/A	~10^2	Very low	Voice only
2G	1990–2000	64 kbps	40 kbps	≈200 ms	95%	10^3	Low	SMS, early M2M
3G	2000–2010	2 Mbps	0.4 Mbps	≈100 ms	99%	10^4	Moderate	Mobile Internet, video
4G (LTE)	2010–2020	1 Gbps	100 Mbps	≈10 ms	99.9%	10^5	High	IoT expansion, broadband
5G (IMT-2020)	2020–2030	20 Gbps (peak), 100 Mbps user	1 Gbps DL	≤1 ms	99.999%	10^6	10× > 4G	Massive IoT, URLLC, eMBB
6G (Target)	≈2030 +	0.1–1 Tbps (projected)	10–100 Gbps	< 0.1 ms	> 99.9999%	> 10^7	> 100× > 5G	Intelligent IoT, Internet of Senses

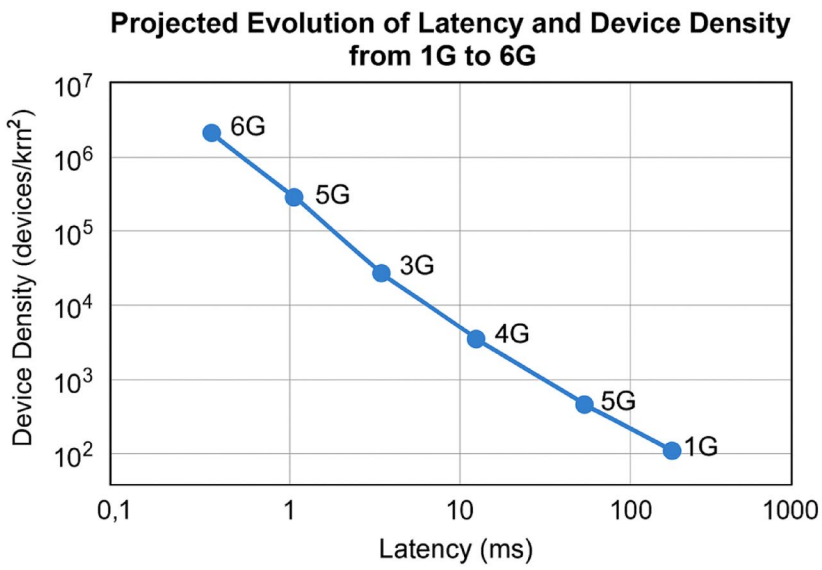

Fig. 1 Projected evolution of latency and device density from 1G to 6G

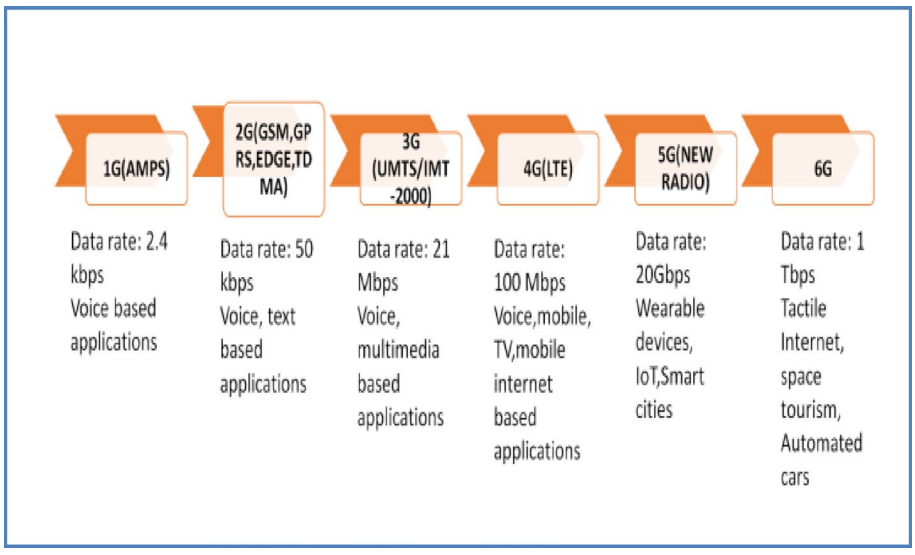

Fig. 2 Unveiling the next frontier: The evolutionary trajectory of 6G technology

2.1 Heterogeneous access and unlicensed spectrum for IoT

Although 2G cellular networks (GSM, IS-95, PDC) were mainly used to offer circuit-switched voice and low-rate data on licensed spectrum, IEEE 802.11 Wi-Fi family, on the other hand, developed independently in unlicensed frequencies to enable high-throughput local wireless access.

Wi-Fi is not included in the evolution of the cellular technology, but it has become an important low-cost complementary technology in the IoT, providing a low-range and inexpensive technology that can be installed at home, on a campus, or industrial location.

Heterogeneous multi-radio access is anticipated to be integrated in modern 6G-IoT architecture with dynamic combinations of licensed 3GPP spectrum with unlicensed

Wi-Fi 6/7, Bluetooth LE and LoRa systems to maximize energy consumption, cost and coverage.

3 Vision of 6G for IoT networks

6G networks are expected to be intelligent, AI-native communication systems that can combine satellite, aerial, marine, and terrestrial domains with ease. 6G will offer genuinely immersive and adaptive connectivity for extensive Internet of Things (IoT) ecosystems by combining communication, sensing, and computation [4, 5].

6G seeks to provide ubiquitous intelligence, semantic communication, and sustainable operation across billions of devices, in contrast to 5G, which primarily concentrates on improved broadband and low-latency services. Future 6G-IoT systems will support a variety of use cases, including digital twins, autonomous vehicles, smart cities, remote healthcare, and tactile Internet experiences, through terahertz (THz) and optical links, AI-driven automation, and reconfigurable intelligent surfaces (RIS) [6, 12].

Figure 3 illustrates the vision of a 6G-IoT ecosystem — integrating terrestrial, aerial, and satellite infrastructures to create a unified, intelligent, and global communication fabric. The vision for 6G-based IoT goes beyond traditional land-based communication to include connectivity across land, air, and sea. These expectations align with the ITU-R and 6G Flagship frameworks, which define intelligence-centric IoT ecosystems as the foundation of IMT-2030 [134, 135].

6G will use cutting-edge technologies like terahertz (THz) links, reconfigurable intelligent surfaces (RIS), edge computing, and artificial intelligence (AI) to make a single

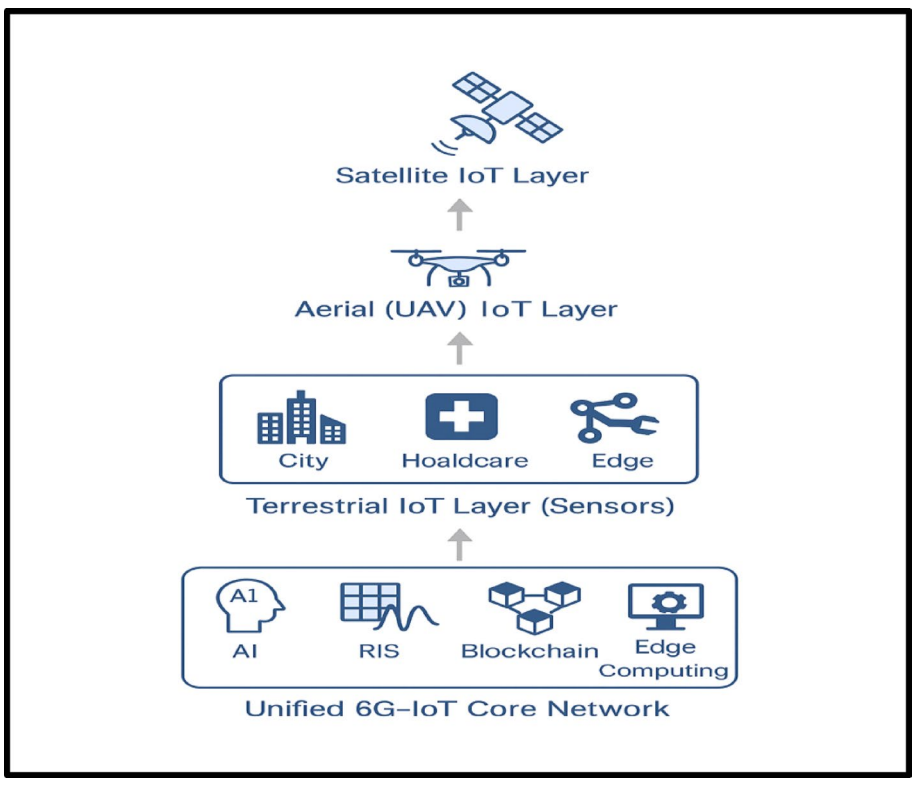

Fig. 3 Vision of 6G-IoT: integrated terrestrial–aerial–satellite architecture enabling intelligent and sustainable IoT applications

communication network that can provide smart, flexible, and energy-efficient IoT services all over the world.

Figure 4 shows this vision by showing how 6G-IoT applications are spread out across different areas:

- Smart cities, self-driving cars, healthcare IoT (HIoT), industrial IoT (IIoT), and agriculture IoT are all connected through edge computing and RIS-assisted communication.
- Aerial IoT: Using drones and UAVs to collect data, keep an eye on traffic, manage disasters, and sense the environment.

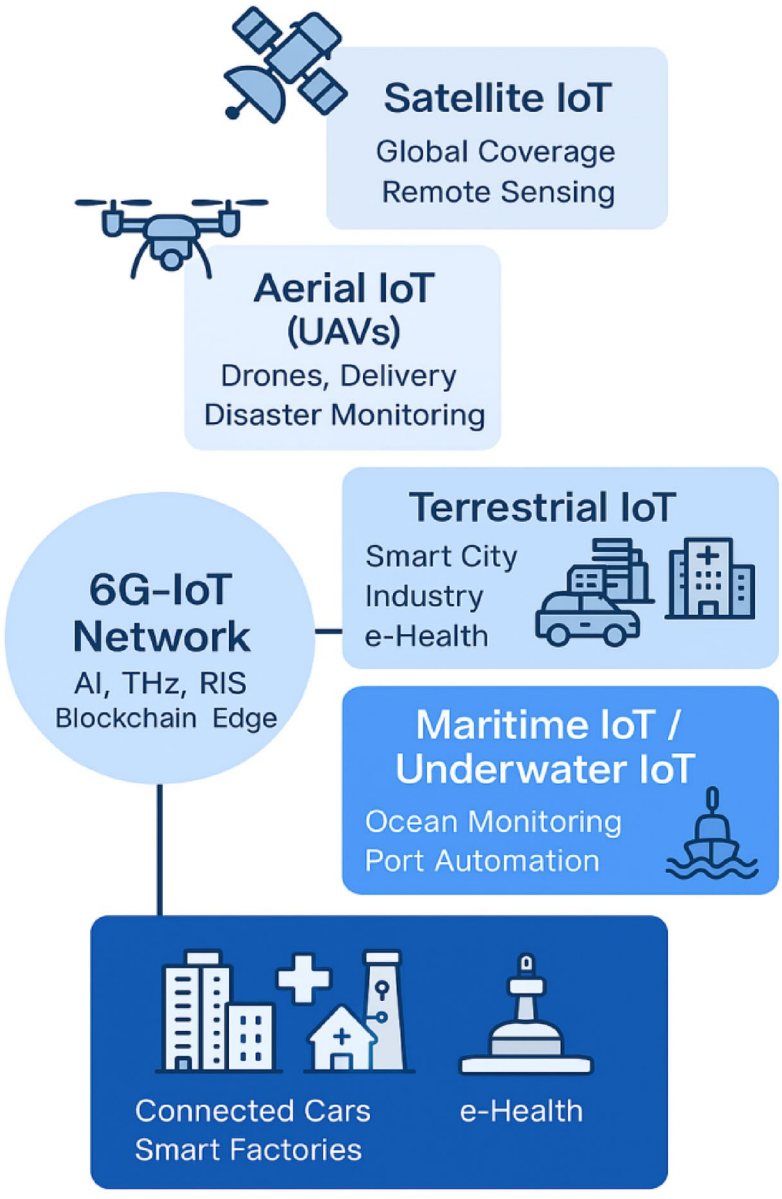

Fig. 4 Future 6G-IoT application ecosystem integrating terrestrial, aerial, maritime, and satellite networks with intelligent coordination

- Maritime/Underwater IoT: Keeping an eye on the ocean, finding your way around, automating ports, and gathering data from below the surface.
- Satellite IoT: covers the whole world for remote sensing, cross-border logistics, and emergency communication.

The 6G-IoT ecosystem will be a smart network with many layers, where land, air, and satellite infrastructures work together perfectly to provide IoT services that are very reliable and aware of their surroundings.

3.1 A literature survey on 6G and IoT

Considerable research has been done in recent years to investigate 6G technologies and the mechanisms that enable them. Numerous studies [12–15] looked at various networking scenarios, such as 3D radio connectivity, decentralized networking, and cellless architecture. While [14] provided a historical overview of wireless systems from 1G to 6G [12] and [13], introduced predictive frameworks. In [15], the topic of intelligent surfaces' function in future networks was covered.

[16–24] gave thorough explanations of 6G enabling technologies and use cases, such as device-to-device communication [27], mmWave-enabled satellite integration [26], blockchain for privacy [25], and optical communication [5]. In [9, 28–30], the challenges of integrating artificial intelligence (AI) into next-generation networks are described.

Additional research [31–33] examined blockchain and AI for resource management, and [33, 34] looked into the use of spectrum in the mmWave and THz bands. Ultra-precise localization and sensing through advanced beamforming and imaging technologies is another anticipated benefit of 6G [35, 36]. Visible light communication (VLC), ML-based communication, and applications like holography, autonomous driving, and industrial IoT were covered in recent surveys [37, 38, 39].

Although these studies offer insightful information, the majority concentrate on generic 6G communication technologies. There is currently no comprehensive review that incorporates these findings into the framework of the Internet of Things, emphasizing the enabling technologies, architectures, and difficulties unique to 6G-IoT.

Table 2 illustrates this gap by contrasting previous survey papers and highlighting this work's unique contribution.

3.2 Summary

This section reviewed previous 6G-related literature and presented the 6G vision for IoT networks. While previous works have looked at different aspects of 6G, a thorough and IoT-centered analysis of 6G systems—covering enabling technologies, application scenarios, and research challenges—remains an open area, as shown by the survey comparison in Table 2. Recent reviews have also highlighted the critical importance of THz communication and AI-native orchestration for 6G-IoT integration [130, 132].

The following sections of this paper fill that gap.

4 Driving force to mode towards 6G

The European Commission is increasing funding for the Gigabit Connectivity project with the goal of influencing Europe's digital destiny [7]. In October 2020, next generation mobile networks (NGMN) unveiled their idea for 6G. In order to further the growth

Table 2 Comparison of existing 6G surveys and this work

Reference	Year	Focus/scope	Main contribution	Limitation/gap	This work's distinction
Giordani et al. [26]	2020	Overview of 6G architecture and use cases	Introduced early vision for 6G communication systems	Focused on general architecture, not IoT-specific	Our work extends toward IoT integration and application mapping
Chowdhury et al. [27]	2020	THz and optical wireless technologies for 6G	Detailed PHY-level enablers	Lacks network-level or IoT-centric view	We emphasize IoT-driven system integration using these enablers
Strinati et al. [28]	2021	AI and edge intelligence for 6G networks	Discussed intelligent networking concepts	No detailed IoT taxonomy or application focus	We combine AI, IoT, and system-level design
Dang et al. [29]	2020	Comprehensive 6G survey (use cases, technologies, trends)	Broad system overview	No dedicated IoT-centric analysis	We focus exclusively on 6G-IoT and cross-domain integration
Saad et al. [30]	2019	Vision of 6G wireless systems	Conceptual vision paper	Predates IoT emphasis; lacks recent enabling technologies	We update scope to 6G-IoT using latest literature
You et al. [31]	2021	Integrated terrestrial–aerial–satellite networks	Proposed space-air-ground integration framework	Did not explore IoT-specific design	Our work links space-air-ground integration with IoT applications
This Paper	**2025**	**6G-enabled IoT architectures, technologies, and challenges**	**Unified review of 6G-IoT systems, applications, and research challenges**	—	**First comprehensive survey focusing specifically on 6G-IoT**

of international mobile telecommunications (IMT), ITU-R decided in February 2020 to start researching potential future technology advancements (Series, 2015).

The IMT-2020 Standard criteria for 5G network standards were defined in 2015 by the ITU radiocommunication sector (ITU-R). The 3GPP released R13 for 5G specifications at the same time. Estimates suggest that ITU will finish standardizing 6G (ITU-R IMT-2030) by the end of 2030, and 3GPP will finish standardizing 6G in R23 [5]. In July 2018, the ITU formed a focus workgroup to investigate B5G/6G system technologies [7]. 2018 saw the launch of the flagship initiative 6Genesis by the Academy of Finland, which focused on 6G technologies [8]. In a similar vein, B5G/6G communication technology research projects have been launched by nations like China, the US, South Korea, Japan, and Russia [4, 5, 9–11].

4.1 Limitation of 6G

Although URLLC can be supported by 5G communication systems, its limits derive from the short-packet, sensing-based URLLC features that are built into 5G. These limitations make it more difficult to provide services that need to be highly reliable, low-latency, and concurrently large data rates. As a result, applications with high requirements, including multisensory XR- AR, mixed reality (MR), and VR apps, are unable to fully utilize 5G cellular infrastructure. XR technologies are developing quickly and are being used in a wide range of sectors, such as industry, education, healthcare, and entertainment. They provide consumers with fresh methods to engage with digital information and immerse themselves in situations that blur the lines between the real and virtual worlds.

Furthermore, future internet of everything (IoE) applications require the convergence of communication, control, computation, and sensing functions. But the 5G infrastructure has mainly disregarded this important factor. The network must provide

heterogeneous devices with high data rates, dependability, and low latency for both uplink and downlink communications in order for IoE services like autonomous systems to function properly [8, 97].

Developing a disruptive 6G wireless network is necessary to overcome these limitations. A network of this kind needs to be carefully planned, considering the unique needs of new IoE services as well as any potential technology advances that may be related. In the rapidly changing world of wireless communication, this strategy is crucial to the proper functioning of IoE services, including autonomous systems.

5 Performance metrics, application scenarios, and example industry verticals

Performance metrics for 6G wireless communication technology are expected to soar to previously unheard-of heights, with targets of data rates that are in the thousands of gigabits per second range, ultra-low latency in the microsecond or nanosecond range, improved energy efficiency, increased coverage, and increased capacity. These developments open up revolutionary application scenarios in a variety of industries. High-speed, low-latency connectivity can be advantageous for immersive experiences like augmented reality (AR), VR, and MR. It can provide smooth interactive gaming, immersive education, and real-time distant collaboration. In a variety of industries, autonomous systems—such as cars, drones, and robots- use ultra-reliable communication to enable accurate navigation, monitoring, and control. 6G-enabled systems are being used by sectors such as manufacturing, healthcare, and smart infrastructure in order to improve sustainability, efficiency, and productivity through real-time monitoring. Various scenarios are discussed in the following subsections.

5.1 Performance metrics and application scenarios

Although 5G networks attain a maximum data throughput of 20Gbps, 6G networks are expected to use THz and optical frequency bands to reach 1–10Tbps. Particularly with these high-frequency bands, the user-experienced data rate can reach Gbps levels, and the area traffic capacity can exceed 1Gbps/m^2. In order to offset a 100-fold increase in data throughput, spectrum efficiency is expected to grow by 3 to 5 times, and network energy efficiency must increase by more than 100 times when compared to 5G. The incorporation of AI for improved network automation and management can enable this breakthrough.

In addition, it is anticipated that the connection density would increase by a factor of 10–100 due to the influence of highly diversified networks, a wide range of communication scenarios, an abundance of antennae, and large bandwidths. This rise will be facilitated by a variety of mobility types brought about by satellites, unmanned aerial vehicles, and ultra-high-speed trains that can reach speeds of over 500 km/h. Less than 1ms of latency is the goal for certain applications, and it is necessary to include other crucial criteria like cost effectiveness, security capacity, coverage, and intelligence level [40].

For comparison, in March 2019, Bell Laboratories proposed key performance metrics that included: a peak data rate of more than 100Gbps, a user-experienced data rate of 1-10Gbps, 107devices/km^2 for connection density, latency of less than 0.3ms, energy efficiency ten times higher than 5G, and capacity ten times higher than 5G systems. Additional metrics include six “9” reliability, 1 m level outdoor positioning accuracy, and 10 cm level inside positioning accuracy. Key performance metrics like > 1Tbps peak

data rate, 1Gbps user-experienced data rate, 10–100μs latency, 1000 km/h mobility, 107devices/km² connection density, 1Gbps/m² area traffic capacity, and 10–100 times energy efficiency and 5–10 times spectrum efficiency compared to 5G are among the things the authors of [41] proposed. While [42] offers a similar perspective on 6G essential performance indicators, as noted in Section I, none of these papers have yet considered the particular traffic and performance requirements needed to support distributed AI in wireless network control systems for haptic internet applications as shown in Fig. 5.

Although 5G is mostly focused on eMBB, mMTC, and uRLLC, 6G networks are expected to have a much wider reach, which will increase the number of possible application scenarios. The authors of [41] presented the enhanced situations as enhanced-uRLLC (euRLLC), umMTC, and feMBB. There was also other noteworthy application situations found, like LDHMC and ELPC. These were referred to as ultra-high speed with low latency communications (uHSLLC), ultra-high data density (uHDD), and ubiquitous mobile ultra-broadband (uMUB) in [42, 43]. went on to describe further uses, such as super IoT, AI, and mobile ultra-wideband.

In February 2020, the ITU released the first 6G research timetable. The study of technology trends and vision is anticipated to be finished by 2023. Founded in July 2018, the ITU-T Focus Group Technologies for Network 2030 (FG NET-2030) seeks to investigate network capabilities for 2030 and beyond. It foresees meeting cutting-edge needs in emerging vertical industries for high-precision communication, quick reaction in emergency circumstances, and holographic communications. The comparison of 5G and 6G is shown in Table 3.

The human-centric communication paradigm has a wide range of possible applications in the 2030s. This paradigm has the potential to improve vehicle communication systems

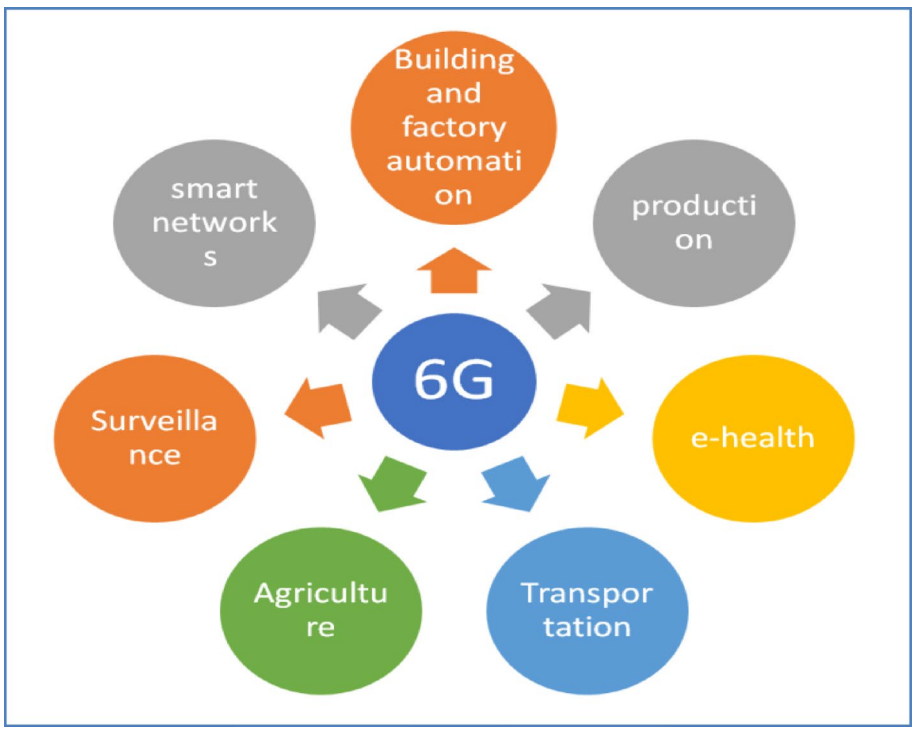

Fig. 5 Emerging horizons: Diverse applications of 6G technology in a hyperconnected world

Table 3 Comparison use cases of 5G and 6G

Use case	5G	6G
Centre of gravity	User-centric	Service-centric
Augmented reality for industry	Low resolution and high-level tasks	High resolution with multi-sensing and comprehensive level tasks
Tele-presence	Limited scale and high video quality	Mixed reality
Security surveillance, detection of defects	External sensing with limited automation	Fully automated through integrated radio sensing
Dynamic digital twins and virtual worlds	No	Yes
Data center wireless	No	Yes
Automation, distributed computing	Microsecond level tasks	High precision tasks at nano-second level
Ultra-sensitive applications	Not feasible	Feasible
Zero energy devices	No	Yes
Groups of robots or drones	Might be 'Yes'	Yes
Bio-sensors and AI	Limited	Yes
True AI	Absent	Present
Reliability	Not extreme	Extreme
VAR (Value-Added Reseller)	Partial	Massive scale
Time buffer	Not real-time	Real-time
Capacity	1-D (bps/Hz) or 2-D (bps/Hz/m^2)	3-D (bps/Hz/m^3)
VLC (Visible Light Communication)	No	Yes
Satellite integration	No	Yes
WPT (Wireless Power Transfer)	No	Yes
Smart city components	Separate	Integrated
Autonomous V2X (Vehicle-to-Everything)	Partially	Fully

in the context of smart automobiles, leading to more user-friendly and engaging experiences that increase convenience and safety. Advanced human-machine interactions in smart manufacturing will streamline production procedures, boosting productivity and customizability. Human-centric technology will be seamlessly integrated into Industry 4.0, allowing smarter and more responsive manufacturing settings. Furthermore, more immersive and customized user experiences will be provided via wearable technologies and integrated headsets, completely changing how people interact with their gadgets. Healthcare will be greatly impacted by implanted sensors, which will offer continuous health monitoring and real-time data transmission [43]. Strict limitations on transmit power and frequency ranges are especially necessary when it comes to wearable health devices because of their interaction with the human body and the sensitive data they gather [43].

To sum up, the three 5G scenarios (feMBB, umMTC, muRLLC, MBRLLC, and ERLLC) have been improved upon for the 6G application scenarios, showing notable gains in performance. Furthermore, a multitude of new application scenarios are made possible by 6G, such as the internet of bio-nano-things, distributed AI (e.g., federated learning) applications, remote holographic unmanned systems, human-centric services, long-distance and high-mobility communications, extremely low-power communications, convergence of communications, computing, control, localization, and sensing, and space-air-ground-sea integrated networks [44]. A comparison of the requirements for key performance metrics and application scenarios between 5G and 6G is shown in Table II.

5.2 Example industry verticals

The potential of the current 5G wireless communication networks to act as the fundamental framework for a contemporary information society has been shown. Based on 5G networks, a number of industry verticals have evolved, and it is anticipated that these will continue to grow in 6G networks. We specifically highlight some example industry verticals, such as digital twin body area networks, cloud-based VR, IoT industrial automation, cellular vehicle-to-everything (C-V2X) communication, and wireless network control systems intended for energy efficiency and federated learning applications.

5.2.1 Cloud-based virtual reality

The military, healthcare, and education are just a few of the professional fields that have used VR technology in recent years [45–48]. Because VR experiences are immersive and interactive, they have the potential to overcome risk and expense barriers that exist in the real world, revolutionizing both our work and daily lives. Since VR generates computer-generated surroundings, high processing power is essential to provide immersive experiences. Because of its cost, portability, and small size, mobile virtual reality is expected to be the way of the future. Unfortunately, the high-fidelity VR requirements are not met by the processing power of existing mobile VR systems, leading to an inadequate quality of experience (QoE). Three conditions must all be met at the same time by the perfect VR system: portability, high-quality graphics, and cost.

With the development of 5G and multi-access edge computing (MAEC) technologies, cloud-based VR is now a feasible goal [49]. Since 2018, telecom companies like China Mobile, LG Uplus, and China Telecom have begun to provide cloud VR services. Cloud VR is becoming more and more popular because of its many advantages for users, developers, and operators. Without the need for expensive terminals, cloud VR gives users access to top-notch VR services whenever and wherever they want it. By reaching a large consumer base and lowering porting expenses, developers can take advantage of new revenue prospects. Cloud VR presents a compelling use case for operators to take use of 5G and MAEC, enabling the digital transformation of their business from connectivity to cloud platforms and from traffic operation to experience operation.

Small-scale solutions are currently available, but as the 6G era draws near, there are a number of important issues that must be resolved. One of the most important features is that cloud VR platforms rely on costly accelerators for mobile edge computing (MEC), which drives up investment prices. Furthermore, very low latency is required for cloud VR services to provide a better user experience. Finally, the research community has the chance to optimize the processing pipeline thanks to the cloud VR architecture. To overcome these obstacles, a thorough examination of current scholarly and commercial research and development is needed to improve the cloud VR system in advance of 6G [50].

In a commercial deployment context, rendering services for numerous VR apps must be provided concurrently by a group of graphics processing units. For efficient operation, different technologies must be integrated to reduce investment expenses. Rendering performance is improved when foveated rendering is used. Foveated rendering helps to minimize transmission bandwidth and cloud computing burdens in the context of cloud VR.

The method presented in [50] uses eye-tracking to reduce shading, demonstrating how rendering performance is improved through foveated rendering, which limits the field of vision. The cloud VR system that the authors introduced in [51] is built on the simulator platform SIGVerse and uses immersive interfaces to integrate human gaze controls and interpersonal interactions. The results of the experiments demonstrate how well the cloud VR system with foveated rendering works. In [52], researchers evaluated DeepFovea experimentally. DeepFovea is an AI-assisted foveated rendering technique that achieves over 14 times compression on RGB video with negligible loss in perceived quality. The rendering demand is considerably decreased by deep learning-based foveation, according to the results.

In a cloud VR system, lowering latency is a crucial technical obstacle to delivering the most immersive experience possible. Frames are captured and encoded on edge cloud servers following computation, and then they are sent over the 5G downlink network. In cloud VR, low-latency video compression techniques are proposed to improve client experience and efficiency. In order to speed up the encoding process, writers in [53] proposed an improved codec that used rendering information to lower the complexity of encoding. According to experimental data, encoding time can be reduced by 42% without sacrificing graphic quality. The authors of [54] investigated frame interpolation by incorporating motion estimation, utilizing a specialized interpolation method for accurate interpolation, and image homography for better motion prediction. According to experimental results, the entire encoding process was sped up by 18%.

Different rendering pipelines are involved in the shift from local to cloud rendering, which is comparable to the evolutionary routes observed in offline rendering pipelines for film production. Application, geometry processing, rasterization, and pixel processing are the four conceptual phases that make up the local rendering process. On the other hand, there are four more steps in the cloud rendering process: application, rasterization, pixel processing, geometry processing, frame capture and encoding, frame transmission, and frame decoding and processing. The bottleneck, which establishes the limit for the entire system, is the slowest process in the pipeline. It's critical to optimize the cloud rendering pipeline's performance. Authors in [55] created a VR framework that divides background environments and foreground interactions for every frame. To lessen server computing burden, they used split rendering architecture on both the client and the server. The split architecture satisfies latency and performance criteria, as shown by the experimental findings. A modified planar map rendering process was suggested by the authors in [56] in order to improve efficiency and lower latency.

5.2.2 Automation of the IoT industry

The fourth industrial revolution stage, known as "industry 4.0," seeks to improve industrial output by utilizing cyber-physical systems to greatly increase flexibility, variety, usability, and efficiency. Industry 4.0 introduces information and communication technology (ICT) approaches, in contrast to the third stage, which concentrated on factory automation using electronics and information technology (IT). Using the IoT to take use of pervasive connection services in the industrial manufacturing sector is one example of this [57]. 5G's multifaceted features, especially its focus on machine-to-machine and ultra-low latency/high reliability connectivity, make it a perfect "one size fits all" option for Industry 4.0.

It is necessary to address the unique issues faced by this vertical industry, nevertheless. These hurdles include a variety of use cases with demanding network performance standards, intricate propagation environments that may be highly interfered with, easy integration with current installations, safety and security concerns, and the nuances of terminology used in different industries. The 5G alliance for connected industries and automation (5G-ACIA) was founded in April 2018 to solve these issues and promote cooperation across industries. It functions as an international platform for assessing the technical, regulatory, and business issues of 5G in the industrial sphere, considering the full ecosystem and pertinent stakeholders including academic institutions, operating industry participants, and members of the ICT sector [58]. In order to use 5G technology for automation in the manufacturing domain, the 5G-ACIA has established particular application areas and use cases that are:

Areas of Application:

- Automated factories
- Automation of processes
- Production and human-machine interactions IT Storage and Transportation
- Predictive maintenance and observation

Common Use Cases:

- Control of motion
- Control over control
- Transportable control panels
- Robots on wheels
- vast wireless sensor networks
- Remote upkeep and access
- AR, or augmented reality
- Process control in closed loop
- Process observation
- Asset management for plants

The 3GPP recognizes the 5G-ACIA definitions as the driving force behind 5G system design to support vertical industries, and these use cases may be relevant to one or more application areas [59]. Outlining specific needs is just as important as identifying use cases—especially the more difficult ones. A few common use cases with explicit performance criteria are shown in Table I given in [58]. Motion control is one of the most rigorous closed-loop control use cases in industrial automation, necessitating deterministic communication capabilities, ultra-high dependability, and minimal latency. Certain use cases involving motion control could present difficulties for 5G, which means they might be investigated early in the 6G development process.

5.2.3 C-V2X V2X: Empowering autonomous vehicles

Vehicle to everything (V2X) is a critical enabler for self-driving cars and offers significant potential benefits in terms of improving traffic efficiency and road safety [60]. In order to provide communication capabilities with low latency, high reliability, and significant throughput across a variety of communication scenarios, including vehicle-to-vehicle

(V2V), vehicle-to-pedestrian, vehicle-to-infrastructure, and vehicle-to-cloud, 3GPP developed the standardized V2X solution known as cellular V2X (C-V2X).

LTE vehicle (LTE-V) and new radio vehicle (NR-V) are two noteworthy C-V2X revisions [61, 62]. Based on the LTE technology, LTE-V is primarily concerned with providing necessary information for fundamental road safety. Conversely, NR-V improves upon the 5G NR system to fulfill more demanding throughput, latency, and reliability requirements for sophisticated V2X applications. These include remote driving, enhanced sensor functionality, platooning, and high-level autonomous driving. NR-V aims for impressive targets: 3ms or less of latency, 99.999% dependability, and up to 1Gbps of data throughput for extended sensor information. NR-V depends on important technologies, namely robust MEC and efficient wireless resource scheduling, to satisfy these KPIs. To maximize resource scheduling on the PC5 interface, a variety of techniques have been put forth [63–65]. The MEC architecture has been well specified by the 3GPP V2X applications. MEC systems, or servers, can be placed strategically inside a cellular network or on the side of the road with a connection to the network. The decision is based on the particular business model and applicable laws. Cooperative perception, cooperative computation, and cooperative decision-making for autonomous vehicles can be supported by C-V2X by utilizing MEC. A wide range of approaches to computing task offloading have been thoroughly investigated in order to manage distributed sensor data computation.

The functionality and intelligence level of autonomous vehicles are expected to significantly advance as AI becomes more widely integrated. To support these developments, the next iteration of C-V2X must provide a complete solution. The essential to the advancement and implementation of connected car technologies, guaranteeing their usability and safety for broad applications were described below.

a) **Air Interface and C-V2X Architecture according to ICDT**: Utilizing information, communication, and data technology (ICDT), the architecture of C-V2X is distinguished by a cellular network topology that integrates completely virtualized network resources, including computation, storage, and communication resources. In order to provide flexible scheduling for C-V2X communication, processing, and data storage and to adjust to the real-time demands of driving scenarios, these resources are dynamically allocated. For C-V2X applications, control nodes and compute nodes—which are made up of central and distributed nodes—are essential for controlling network resources, information, and capabilities on a global scale. Multiple functions, like as sensing, synchronization, and positioning, are required of access nodes in order to facilitate cooperative perception for vehicles with a variety of sensors. With features like massive MIMO, mmWave bands, visible light communications (VLC), and THz communications expected to facilitate cooperation across intelligent transportation system (ITS) entities, high-precision location becomes a crucial problem for C-V2X. It is intended for the C-V2X air interface of the future to use a single wireless waveform for ease of use [66].
b) **Technical Validation and Examination of C-V2X**: C-V2X testing has been conducted by numerous international studies; yet, evaluating C-V2X features and performance in real-world testing scenarios is still difficult. For C-V2X-based autonomous driving testing, the idea of a digital twin—which provides virtual representations of systems throughout their lifecycle—presents a potential approach. Real-time autonomous

vehicle testing in a virtual complicated road environment within a simulated test environment is made possible by the use of digital twin mapping. Using real-time cooperative perception enabled by cloud and edge computing, autonomous cars create a virtual driving environment that serves as the basis for all driving behavior decisions. This virtual driving environment is essentially a digital twin of the real road environment.

c) **C-V2X New Spectrum Considerations**: A study is conducted in [67] to examine the spectrum needs for C-V2X sidelink communications in autonomous driving situations. The groupcast mode in a NR-V system helps with negotiation and decision messages during autonomous driving, while the broadcast mode provides status and environmental information. It is predicted that broadcasting requires at least 30–40 MHz of spectrum, with very little needed for groupcasting. Massive MIMO, mmWave bands, VLC, and THz communications are examples of emerging technologies that provide C-V2X a wealth of new spectrum resources. However, further research is needed to determine the spectrum needs for these possible C-V2X technologies, which means that a sound methodology and assumptions are needed to make reliable judgments.

5.3 Network of digital twin body area

From 2016 to 2018, Gartner regularly ranked the digital twin as one of the top ten strategic technology trends, recognizing its potential for disruptive innovation. Large corporations that have aggressively embraced digital twin technologies include Siemens, Dassault, PCT, and ESI. Developing dynamic monitoring and simulation of the human body's working mechanisms, medical devices, and applications in drug research, epidemic prediction, and tracking are noteworthy endeavors undertaken by firms like Dassault and Hisense. The integration of devices, senses, motion control, and health monitoring within the connected digital and physical spheres was highlighted during the 2019 6G conference.

At the moment, digital technology is mostly concerned with keeping an eye on macrophysical indicators and mitigating common ailments. Still, real-time precision has to be enhanced. Combined with developments in materials science, bioscience, and bioelectronics medicine, 6G technology could lead to the creation of a comprehensive "digital twin of the human body." This idea, which envisions a vast deployment of intelligent sensors (> 100 devices/person) throughout the human body, seeks to accurately monitor a variety of bodily systems, including the neurological, respiratory, and urinary systems, musculoskeletal systems, emotional states, and more. By generating an exact virtual representation of the full human body in real-time, "image mapping" will make individualized health data monitoring possible. When combined with biochemical tests and medical imaging data, AI technology allows for more accurate health evaluations, prompt interventions, and critical information for precise diagnosis and individualized treatment plans.

By simulating a virtual human body using 6G and ICT, the digital twin body area network (DTBAN) makes disease prediction, all-weather tracking, and virtual operation simulations possible. The COVID-19 pandemic serves as a prime example of how important the network's cloud, fog, and sensor layers are for epidemic management. In

the realm of healthcare innovation, a cutting-edge approach involves the utilization of digital twin networks interfaced with advanced computing technologies. Such as:

- Real-time health data is sent from a digital twin network to the edge computing personal network slice.
- Fog computing and secure multi-party computing analyze deployment data to spot irregularities.
- Through the non-contact detection of digital twins, doctors are able to pinpoint the source of infection and facilitate crowd behavior tracking.
- Digital twins speed up data processing by expediting the survey, modeling, and verification of clinical medications.
- The synesthesia network helps patients and their families communicate emotionally while they are alone.

The DTBAN's network design demonstrates the following unique features:

- Compact and adaptable, it moves from the enterprise- or park-level edge computing of the 5G era to the individual-level granularity of the 6G era.
- By choosing and working together across several networks, fog computing and intelligent cooperation allow for autonomous network management.
- Digital twins, created after software defined networking (SDN), greatly expand uplink data proportions by adopting an open AI crowdsourcing paradigm, increasing the value of both company and personal data.

Prioritizing privacy and security, the network employs secure multi-party computing, blockchain, and ensures that privacy data remains within the body network, with only the model exiting, thereby safeguarding security, privacy, and underscoring the value of data and retained computing resources.

It is anticipated that a 6G network will feature major improvements over earlier generations in terms of its base station, user equipment, surroundings, and core network as shown in Fig. 6(a). It is probable that the cloud-native design of 6G's core network will make use of microservices, virtualization, and containerization to improve resource efficiency, scalability, and agility. Network slicing, which divides the core network into several virtual networks suited to different apps, services, or user groups, will be essential to 6G. By processing and storing data closer to the networking edge, edge computing capabilities can be integrated into the core network to reduce latency and facilitate real-time applications. Figure 6(b) shows a comparison of the 5G and 6G network architectures. Based on intelligence, high computing power, and large capacity, the 5G core network has been updated to the 6G core network. An analogous upgrade is made to the access network by integrating satellites, UAVs, and base stations (BSs)/access points (Aps). In addition to the horizontal hand-off seen in 5G, there is also a vertical hand-off in 6G. Additionally, a large number of devices on the user plan can receive frequently needed services with less latency and bandwidth consumption thanks to the integrated 6G network infrastructure that includes fog computing and MEC. Massive MIMO technology and sophisticated beamforming techniques will probably be seen in 6G base stations in order to increase spectral efficiency and service. In order to improve connectivity, coverage, and performance, the 6G environment will be typified by intelligent surfaces, decentralized antenna systems, and sensor networks. In order to enable seamless connectivity

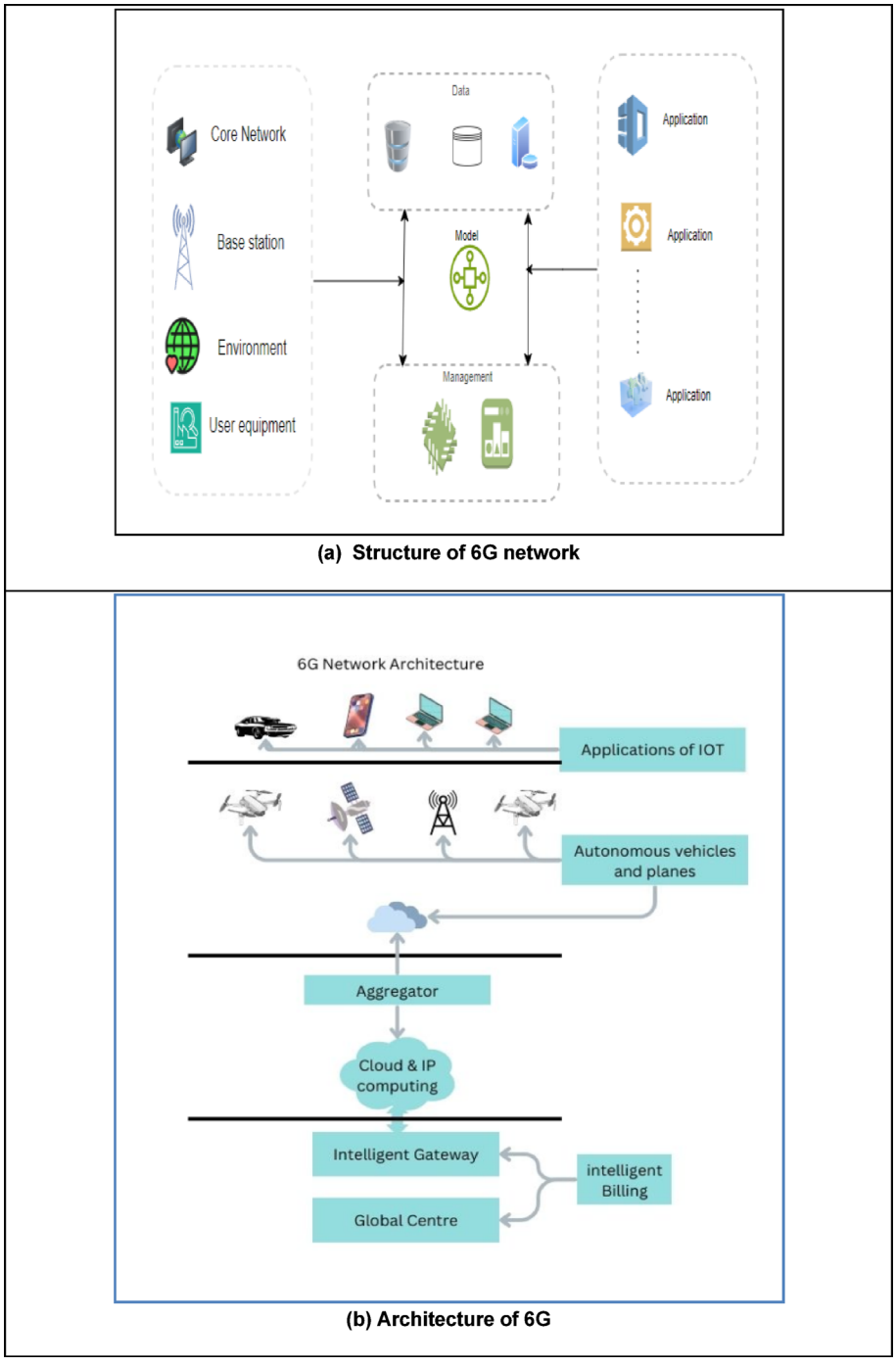

Fig. 6 Comprehensive framework and architecture of 6G: A multi-layered ecosystem enabling next-generation connectivity and services

in a variety of situations, 6G networks will be comprised of diverse network deployments that combine distributed antennas, macrocells, small cells, and other network elements. Intelligent and self-sufficient, 6G user equipment will make use of AI algorithms and machine learning algorithms to optimize power management, resource allocation, and network selection. Phased arrays and adaptive transmitters are examples of cutting-edge antenna technology that user devices will use to enhance signal transmission, reception, and interference rejection.

5.4 Controlling wireless networks with minimal energy and federated learning systems

The modern world is seeing an increase in applications that mostly depend on data correlation for analysis. The fight against pandemics, such as COVID-19, provides an interesting case. Mobile device proximity data is crucial for analysis and subsequent correlation across networks to stop the disease's spread. An additional situation entails automated driving at crosswalks where a variety of users are present, requiring the incorporation of both mobile and stationary sensors like radar and lidar. This combination is essential for comprehending the surroundings and efficiently directing traffic using a decentralized control approach.

The use of mobile terminals for large-scale data analytics and the implementation of tactile internet both highlight the necessity of a 6G network. This cutting-edge network is essential for handling the massive amounts of data traffic needed to put distributed learning algorithms into practice. The latency requirements can be severe (about 1ms) or reasonably low (more than 10ms), depending on the algorithmic distribution. Data volumes that require transportation will exceed the traffic that primarily consists of video and streaming on present 5G networks, reaching previously unheard-of heights.

Increasing the softwarization of network functions is a natural way to get 6G networks ready for these unknown traffic demands. This methodology facilitates adaptability to changing needs. Nevertheless, there is a cost associated with using commercial off-the-shelf (COTS) server technology: higher power consumption. Moore's law plateauing means that an energy-efficient COTS server is not enough to meet the growing need for computing power. For COTS servers, semiconductor scalability is insufficient to offset their higher power consumption. Consequently, the creation of domain-specific compute engines optimized for 6G is currently the only conceivable answer. This method is similar to the one that is currently being used for effective ML deployment, highlighting the necessity of similar solutions for constructing domain-specific 6G machines [68, 69]. 6G wireless communication technology includes a range of innovative characteristics that have the potential to revolutionize technological capabilities and connectivity as shown in Fig. 7. Primarily, ultra-high bandwidth holds the potential to enable blazingly fast data transfer and open up possibilities for flawless streaming of high-definition video, virtual worlds that are lifelike, and unparalleled real-time communication. In addition, there is the goal for ultra-high precision, where new developments in positioning technologies usher in an era of accuracy critical to applications like industrial automation, precision farming, and driverless cars. In addition, 6G aims to provide exceptionally high performance in a number of measures, including data rates, latency, dependability, and connectivity, which will support mission-critical operations, immersive apps, and processing intensive workloads.

6 6G network requirements

As widely reported in the literature, a significant amount of recent research has been devoted to investigating 6G applications, enabling technologies, architectures, and the related research difficulties. A thorough analysis of the applications, enabling technologies, and major research problems in the field of 6G technology was provided by [72]. In addition, they discussed applications, performance indicators, new consumer services in 6G networks, and developing trends in 6G. In [73], the concept of AI-infused 6G wireless networks was presented, providing insight into the architecture of 6G networks and

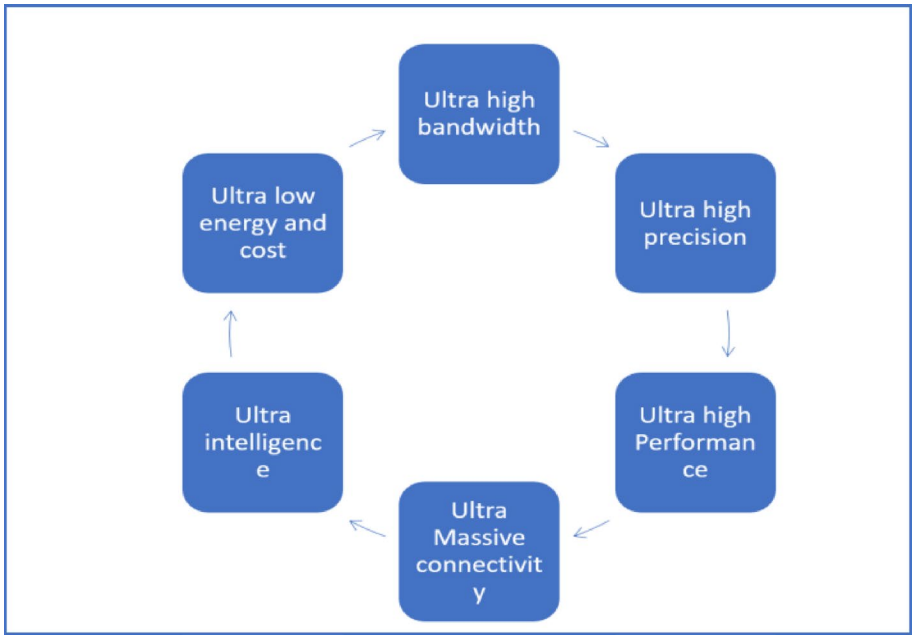

Fig. 7 Unlocking the potential advantages and benefits of 6G technology for future connectivity

their uses in a range of AI-powered smart services. Tariq et al. [74] concentrated on the application of 6G, investigating the technologies that make it possible and tackling related research issues.

Moreover, Giordani et al. highlighted the development of wireless communication systems toward 6G networks and provided a list of particular use cases [75]. The main topics of discussion were the essential 6G enabling technologies, the difficulties they present, and the variety of uses connected to these developments.

Furthermore [71], provides a comprehensive analysis of the basic needs, driving forces, design concepts, and enabling technologies that are essential to the creation of 6G. In addition [76], discusses possible technologies for 6G wireless networks and provides a thorough rundown of the time, frequency, space, and resource usage relevant to 6G networks. The development of 6G wireless networks and the expected difficulties in putting them into practice are also covered in length, including key methodologies. Similarly [77], investigates the characteristics of 6G that concern user-experienced data rates, latency, energy efficiency, and peak data rate. In [78], the difficulties related to the intelligence and different machine learning algorithms included in 6G wireless systems are discussed.

The main 6G enabling technologies—intelligent communication conditions, layered structural design, and open research challenges—are covered in detail by Akyildiz et al. [79]. Furthermore [80], explores the technological advancements, uses, specifications, and general idea of 6G.

6.1 Interconnectivity

With the widespread adoption of seamless and infinite wireless connections, societies are set to become possibly data-driven entities in the near future [85]. Many strategies, such the adoption of new 5G radio technology and the concurrent use of licensed and unlicensed bands, have been considered in order to maximize the usefulness of the 5G

network for a variety of smart applications [86–88]. Though short packets for URLLC and basic smart IoE services are among the expected benefits of 5G networks, they also reveal inherent limitations and complexities in meeting the evolving requirements of future smart city IoE applications [70, 72].

It is clear that 5G networks' capabilities and key performance indicators are unable to meet the growing demands brought on by the growth of data-centric and automated operations [75]. Short packets for URLLC implemented by 5G networks are not generally usable for applications like haptics, linked autonomous vehicles, and telemedicine, which are predicted to rely on long packets with ultra-high reliability and high data speeds [72]. Moreover, the 5G network has drawbacks including low mmWave connection range, signal disruptions, Gbps-level transmission data rates, and restricted coverage in rural and isolated places, in addition to its limited connectivity density (106/km2) [70].

6.2 Latency

One unique aspect of 5G networks is its low latency, which is made possible by deterministic networking (DetNet), which guarantees precise and timely end-to-end latency. It is anticipated that 6G mobile networks will outperform 5G networks in terms of time and phase synchronization precision [70]. This makes 6G appear like a promising technology that will serve a variety of industries and improve people's quality of life and perspective of life in the near future [90]. In addition, 6G networks are expected to surpass the constraints of 5G networks in order to meet the more demanding needs of the upcoming smart systems.

The enhanced features of 6G networks are intended to outperform those of 5G networks. These features include sub-millisecond time synchronization, sub-centimeter geo-location accuracy, nearly 100% geographical coverage, 10 times lower latency, high-level intelligence for full automation, sub-millisecond time synchronization, higher transmission data rates (Tbps), and 100 times higher connection densities [79, 89].

6.3 Reliability

AI will be a key component of 6G networks, which are expected to offer 99.9% reliability and enable the optimization of numerous wireless network difficulties [91, 92]. The introduction of 5G networks has highlighted the trade-offs related to softwarization, in which a virtualized radio access network (RAN)'s usage of COTS servers rather than domain-specific processors results in a notable rise in energy consumption. This calls for actions to improve energy efficiency, especially since 6G networks seek to implement a new computing paradigm that capitalizes on softwarization's advantages without enacting the expenses linked to higher energy usage [89]. Furthermore, it is anticipated that the majority of 6G use cases would develop from newly developed features and improved user experiences in 5G system-based apps, then advance through performance improvements and the addition of new use cases [74].

6.4 Methods of computing

Important computer technologies that improve processing power, lower latency, shorten synchronization times, and strengthen overall network resilience—such as cloud, fog, and edge computing—are essential to distributed computing. These technologies are

expected to provide greater dependability, reduced latency, better system coverage, and larger data rates in addition to resolving the short-packet limitation and other limitations of 5G networks [72]. In addition, in order to meet the changing needs of mobile communication in the upcoming years, a human-centric approach to 6G development is prioritized above machine-, application-, or data-centric approaches [71, 83]. 6G would provide much more capacity than its predecessors, allowing for the transfer of enormous volumes of data at previously unheard-of speeds. This enables seamless streaming of high-resolution video, quicker downloads, and compatibility for bandwidth-intensive apps like augmented and virtual reality. 6G networks will allow ultra-low latency, increased bandwidth, and seamless integration of satellite communications thanks to the core network's incorporation of powerful AI for autonomous management, real-time analytics, and enhanced security. Large antenna arrays, THz frequency bands, and sophisticated beam forming will all be used by the base station (ngNodeB) to give previously unheard-of data rates and capacity, enabling huge device connectivity ranging from driverless cars to smartphones. Advanced adaption algorithms will be required due to the very heterogeneous nature of the environment, where a variety of physical and radio factors will impact the quality of the signal. A vast range of devices with improved augmented reality, holographic communications, and IoT integration capabilities will make up user equipment in the future, and data and management will use AI and machine learning to optimize resource allocation, network performance, and energy efficiency.

6.5 Expansion

The core of 6G wireless networks will be defined by the development of new paradigm shifts. Global coverage via integrated space, ground, air, and marine networks is the goal of these networks. UAVs, satellite communication, and maritime communication are the three main ways that wireless communication networks are expected to reach broad coverage and range [93].

6.6 Speed of data transmission

Exploring different spectrum, such as optical frequency bands, sub-6 GHz, mmWave, and THz, is necessary to improve total data speeds. It is anticipated that the combination of 6G networks with AI and ML technology will allow for extensive application, automation, and management. AI-based methods by dynamically instrumenting networking, caching, and computing resources can greatly improve the performance of next-generation networks.

6.7 Safety

In 6G, it is critical to establish strong network security at both the physical and network layers. The development of 6G networks will be greatly aided by highlighting the importance of industry verticals like cloud VR, IoT, industry automation, C-V2X, area network for the digital twin body, and energy-efficient wireless network control, in addition to federated systems of learning [89]. As such, security requirements for 6G systems are critical.

7 Essential enabling technologies for 6G networks

Mobile networks are evolving by leveraging the benefits that have been carried over from earlier network architectures and adding new benefits that are specifically designed to meet the needs of the modern period [76]. Similarly, the forthcoming 6G network will simultaneously integrate new technology to meet future requirements and will also leverage the benefits of 5G architectures. Therefore, it is hypothesized that a wide range of technologies would be utilized by 6G communication systems; a few of these are explained below:

7.1 Internet of things

The IoT is a concept that centers on connecting everything to the Internet. This creates a networked environment where data processing, communication, and sensing may happen automatically without the need for human intervention. The IoT is essential to the wireless network integration of many electronic gadgets. It can provide advantages to end users by collecting data from ubiquitous mobile devices, such as computers, smartphones, RFIDs, actuators, and sensors [94]. According to Cisco's projections [95], there will be 500 billion IoT devices online by 2030.

Furthermore, an IHS Markit report [96] predicts that the number of globally linked IoT devices will increase by an astounding 12% annually, from about 27 billion in 2017 to 125 billion in 2030. Given this, 6G is positioned to be a key component of future IoT networks and applications, offering all-encompassing wireless coverage and including all necessary features like sensing, transmission, computation, cognitive processing, and completely automated control. In order to facilitate IoT connectivity and service delivery, 6G is anticipated to provide wider coverage and more flexibility than the 5G mobile network [85]. In the 6G context, IoT evolution will emphasize ultra-low-power nodes, large-scale device coordination, and energy-aware connectivity through ambient backscatter and energy-harvesting transceivers.

7.2 Artificial intelligence

Intelligence, a relatively new technology included into these networks through the application of AI, is a crucial component of self-sufficient 6G networks [98]. In contrast to earlier generations, where AI's application was limited, 5G networks experienced some integration. Notably, it is anticipated that 6G networks would fully automate via AI, releasing radio signals' full potential and permitting cognitive radio transformations [73]. Through developments in machine learning and AI processes, the use of AI in 6G networks is predicted to result in highly intelligent networks that improve and streamline real-time data transfer. AI approaches provide a number of advantages, such as heightened productivity, decreased processing lags in communication stages, effective problem-solving, fostering communication inside the brain-computer interface (BCI), and managing network handovers and selection.AI-based communication systems will be supported by technologies like ML, cognitive radio, intelligent structures, networks, devices, meta-materials, and self-sustaining wireless networks [98, 99].

The implementation of AI technology is anticipated to facilitate the realization of many 6G services, such as massive machine type communication (mMTC), ultra-high-definition display (uHDD), and uRLLC. They can now be used for spectrum mining, spectrum mapping, and RF signal processing thanks to recent developments in ML techniques. In

order to shape a cognitive radio system based on photonics, combining ML techniques with photonic technologies would further enhance AI evolution in 6G networks.

The physical layer implements AI-based deep learning encoder-decoder setups for tasks like channel state estimation and automatic modulation classification, while the data link layer and transport layer, respectively, extensively explore deep learning-based resource allocation, intelligent traffic prediction, and control [100]. Predictive analysis can be used to identify the optimal methods for data transmission between end users, which is another benefit of utilizing machine learning and big data [98, 99].

In 6G-IoT networks, AI acts as the primary enabler of network intelligence for billions of low-power sensors. Machine-learning-based optimization dynamically allocates spectrum and energy resources to maintain ultra-low latency and high reliability while prolonging battery life across dense IoT deployments.

7.3 Combining energy transfer and wireless information

Wireless Information and Energy Transfer (WIET), which uses fields and waves similar to those used in wireless communications, is a ground-breaking technique in the field of 6G networks. WIET has the ability to significantly increase the battery charging lifetime of wireless systems and enabling battery-free devices in 6G networks. It is expected to support the creation of smart devices without batteries, enhancing the effectiveness of charging as well as prolonging the life of batteries in wireless networks and other devices [101, 102].

This feature is particularly relevant for 6G-IoT, where battery-free sensors and zero-energy nodes rely on harvested RF energy for operation, enabling long-lived and maintenance-free IoT deployments.

7.4 Mobile edge computing

The first move toward edge computing was taken in the 1990 s when Akamai launched content delivery networks (CDNs), with the goal of improving speed and performance. Cloud computing platforms are added to edge computing, which broadens the CDN notion. Brain et al. (1997) emphasized the significance of edge computing in mobile networks [103, 107, 114]. Although cloud computing became well-known in the middle of the 2000 s and was utilized by Apple and Google as their main mobile device infrastructure, Paramir Bahl and associates presented the core ideas of edge computing back in 2009 [95]. Edge computing is important because it shortens the time it takes to deliver cloud computing services to end consumers. It functions as a quick-responding mechanism that processes information quickly. In order to support the heavy execution of AI algorithms, the upcoming 6G networks will integrate edge computing hardware with current 5G and IoT infrastructures [84]. Wide range of applications were listed in Fig. 8.

For IoT applications requiring real-time analytics—such as industrial automation and vehicular IoT—edge computing brings intelligence closer to devices, reducing latency and bandwidth load while maintaining privacy and energy efficiency.

Because mobility-enhanced edge computing (MEEC) has so many applications in dispersed large-scale clouds, it is therefore expected to be a key component of future 6G systems. AI techniques combined with MEC infrastructures will allow for efficient computing for edge system controls as well as large data analytics [105]. In order to best address the demanding needs of heterogeneous compute, communication, and

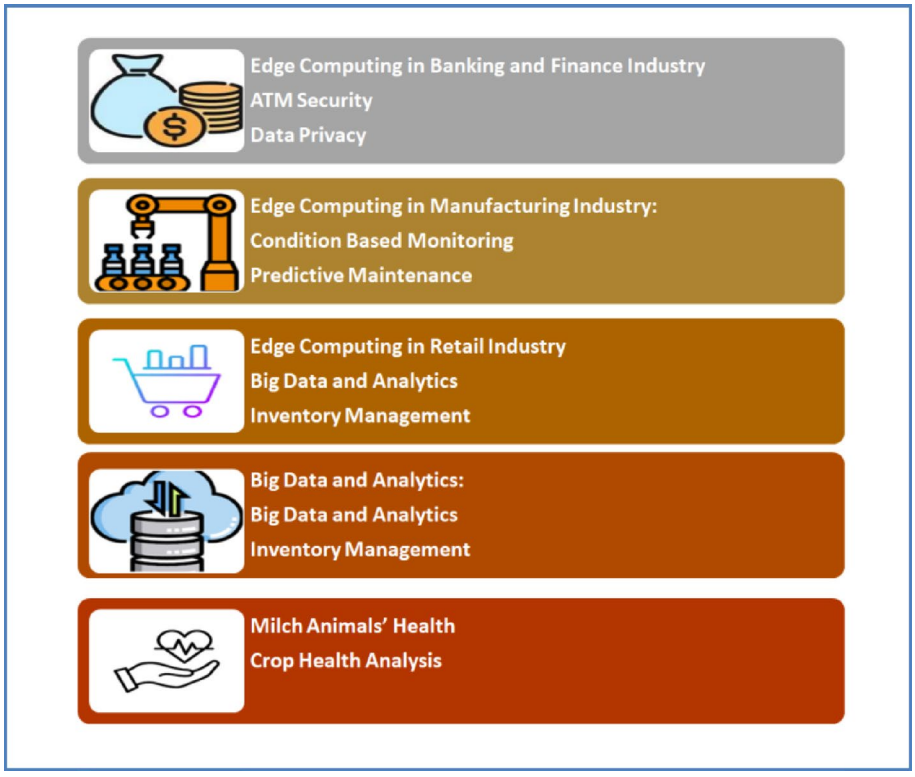

Fig. 8 Empowering the future: Diverse applications of edge computing in real-world scenarios

high-dimensional intelligent configurations in ubiquitous service scenarios, edge-based intelligent computing has arisen [80].

7.4.1 Reporting autonomous vehicle accidents in real time

It is anticipated that the automotive industry would see a significant increase in the number of autonomous vehicles, which will be divided into six categories (0 to 5) according to their degree of automation. Level 5 automobiles are fully automated, with features like lane change and collision avoidance [106]. Level 0 cars have no automation at all. In this situation, handling real-time data from these autos can be greatly aided by roadside units that have edge computing capabilities. Think of a situation where there is a traffic accident and prompt reporting is necessary in order to treat the affected parties with first aid. Effective communication during emergencies is made possible by roadside equipment equipped with edge computing facilities that involve medical teams, road administrators, smart police, and public safety points.By enabling quick first assistance, accidents can be detected and reported in a timely manner, minimizing damage. Safety points with edge computing facilities can automatically detect incidents through algorithms, ensuring fast response and detection of delicate tasks in instances where persons may not be able to report occurrences due to serious injuries.

7.4.2 Intelligent forest fire indication

Inspired by the IoT, the idea of "smart forests" entails using remote sensing to gather environmental data. Early detection is the main goal in controlling wildfires in woods. The deployment of an edge computing-based smart forest fire warning system has demonstrated efficacy in prompting reports of forest fires and facilitating their observation

[98]. Car cameras take photographs all the time, and the server processes them so that they can be used to report fires. Communication might be hampered by slow camera processing, which could result in delayed firefighting reactions. This issue is addressed by edge computing, especially in image processing, which shortens response times and facilitates fast reporting to enable quick decisions regarding fires [108]. Additionally, it is used in a variety of rescue operations via the use of telecommunications [109].

7.4.3 Intelligent parking technology

Living in an urban area means doing a lot of things, including shopping and working, which calls for reasonably priced parking. Through internet connections, users can look up open parking spaces. Conventional parking systems frequently have trouble effectively managing and locating empty spaces. Smart parking systems calculate available parking spaces quickly and offer organized administration by using machine learning algorithms [110]. Smart parking is made possible by edge computing, which uses AI algorithms to find open spaces with the help of image recognition systems. Edge computing is essential for smart parking systems to enable effective and quick empty space identification [111].

7.4.4 Intelligent house

These days, a large number of services are located at the network's edge rather than in the cloud since edge data processing minimizes response times and throughput expenses. This method improves tenants' living comfort and is especially useful for applications such as smart home systems [112]. Smart controls, smart meters, and improved surveillance are just a few of the benefits that come with smart homes. A multi-layer system with decision-making capabilities for home automation is necessary to put these concepts into practice [113]. A crucial part of the multi-layer system, AI algorithms and IoT devices use historical and real-time data to make decisions. These technologies are already present in a number of items, including as washing machines, refrigerators, air conditioners, and smart TVs.

7.5 Combining communication and sensing

The capacity to recognize constantly changing environmental states is essential for self-directed wireless networks because it allows for efficient information sharing between different nodes [115]. In order to achieve autonomous or self-directed systems in 6G networks, sensing and communication must be seamlessly integrated. This is a difficult task that involves a large number of sensing objects, sophisticated communication resources, multi-level computation resources, and multi-level caching resources.

In 6G-IoT systems, joint sensing and communication allow **context-aware operation** of sensors and robots, enabling smart factories and autonomous environments to adapt resource use and reduce redundant transmissions.

7.6 Slicing dynamic network

One particularly important factor that network operators need to consider in order to guarantee the devoted operation of virtual networks and all-encompassing support for the optimal supply of services to a variety of customers, such as automobiles, businesses, and machinery, is dynamic network slicing. Dynamic network slicing is essential

for handling numerous users linked to heterogeneous networks in 5G communication systems. In order to optimize network performance through traffic flow management, coordinated network resource allocation, and centrally controlled dynamic steering, its implementation necessitates the use of software-defined networking and network function virtualization techniques [116].

Dynamic network slicing in 6G-IoT ensures **QoS isolation** among heterogeneous IoT services—such as healthcare, industrial automation, and smart grids—while minimizing energy overhead through adaptive virtualization.

7.7 Beamforming holographically

With the use of an array of antennas, beamforming, a signal processing technology, can precisely guide radio signals to improve network efficiency, coverage, throughput, Signal-to-interference noise ratio (SINR), user tracking, and interference prevention. A novel technique called holographic beamforming (HBF) uses holograms to guide the beam through the antenna and enable effective beam-shape and beam-direction modifications. HBF uses software defined antennas (SDA), which are much smaller, lighter, cheaper, and use less power than conventional MIMO systems. By utilizing SDAs, HBF may overcome significant obstacles in communication system design, including cost, size, weight, and power (C-SWaP), and this will enable extremely flexible and effective signal transmission and reception in 6G networks.HBF's use in 6G networks is beneficial for extremely effective and flexible signal transmission and reception, especially in multi-antenna communication devices. It also plays crucial roles in wireless power transfer, physical layer security, and augmented network coverage scenarios [82].

For dense IoT environments, holographic beamforming provides directed links to low-power devices, improving connectivity without increasing transmit power and supporting energy-constrained IoT networks.

7.8 Analytics for big data

In order to manage data completely, big data analytics entails the complex examination of large datasets that uncover unnoticed relationships, hidden patterns, and customer trends. Big data analytics is used in 6G communication networks to manage enormous volumes of data. Large data quantities, deep learning protocols, and big data analytics combined with 6G networks should greatly enhance the network because of its self-optimization and automation features. One prominent example of how big data and machine learning can be integrated to support predictive analysis in order to identify optimal user data pathways and reduce end-to-end (E2E) delays in 6G networks is E2E delay reduction [82].

Integrating big-data analytics into 6G-IoT enables predictive maintenance and adaptive resource management for large-scale sensor deployments, thus enhancing operational efficiency and reducing network congestion.

7.9 Backscatter communication

By exploiting existing RF signals, such as cellular communications and ambient television, battery-less devices can communicate with one another through ambient backscatter wireless communication. Although modest communication ranges result in appropriate data rates and enable sensor-based transmission of small monitoring signals

with low power consumption, backscatter systems have a significant potential to be used in future 6G networks to provide vast connection. Non-coherent backscatter communications have the potential to optimize resource deployment and improve services in network devices by addressing crucial needs, including accurate phase and channel state, within the networks [81].

Backscatter communication is a cornerstone for ultra-low-energy 6G-IoT, enabling passive tags and sensors to transmit data using existing RF signals—ideal for large-scale environmental and agricultural monitoring.

7.10 Perventive caching

The widespread implementation of small cell networks to improve overall network attributes, such as coverage, capacity, and mobility management, is a major problem for 6G networks. BS experience severe downlink traffic congestion as a result of this deployment. In order to get over these restrictions, proactive caching offers lower access latency and traffic offloading, which improves user experience in the end. For the deployment of 6G networks to be effective, a great deal of study is needed to jointly optimize a number of factors, including as proactive caching, interference control, intelligently coded schemes, and scheduling algorithms [117, 118].

For IoT applications generating repetitive data, proactive caching at edge nodes minimizes latency and energy use, providing faster responses for time-sensitive IoT services such as smart healthcare and connected vehicles.

7.11 Unmanned areal vehicles

Drones, also known as unmanned aerial vehicles (UAVs), are an essential component of 6G networks; they are frequently used to provide fast data speeds and wireless connectivity. UAVs install BS entities, which contribute to cellular communication. Notably, permanent BS infrastructures are unable to match the special qualities displayed by UAVs, which include simple deployment, reliable line-of-sight connectivity, and controlling movement with degrees of freedom [74]. UAVs are excellent at handling emergency and natural catastrophe scenarios, but terrestrial communication networks have practical and financial constraints. As a result, UAVs provide new opportunities for wireless communications, meeting the needs of wireless networks' uMUB, uHSLLC, mMTC, and uHDD [104, 119]. Their wide-ranging uses cover anything from improving network connectivity to duties like monitoring pollution, emergency services, disaster relief, parking lots, accidents, security, and surveillance. UAVs are considered one of the most important technologies for 6G networks because of their adaptability.

The use of UAV-enabled backscatter communication is a noteworthy invention that helps with a variety of communication tasks, including providing ambient power and setting up appropriate channel conditions for distant sensors [120]. Non-coherent detection techniques and UAVs could be used to develop air interfaces that are appropriate for 6G networks. A strong resource allocation process based on deep reinforcement learning can be used to integrate UAVs in the intelligence integration of 6G networks.

In the 6G-IoT paradigm, UAVs function as mobile edge nodes, extending connectivity to remote or disaster-affected IoT areas while offering adaptive energy management through solar-powered or energy-harvesting platforms.

7.12 Terahertz communications

Expanding bandwidths and employing cutting-edge MIMO technology are two ways to improve spectral efficiency [121]. In order to accommodate future communication demands, 6G aims to push the frequency boundaries to THz. 5G communications rely on mmWave frequencies for wide applications and better data rates. The frequency ranges and wavelengths of THz waves, also known as sub-millimeter radiations, are 0.1–10THz and 0.03–3 mm, respectively. When the RF spectrum is no longer sufficient to meet 6G network requirements, the THz band, which spans from 275 GHz to 3THz, becomes increasingly important in the context of 6G. The present mmWave spectrum (30–300 GHz) can be combined with the THz range (275 GHz–3THz) to greatly boost the capacity of 6G networks. Experimental analyses confirm that combined THz–mmWave operation enhances IoT backhaul performance and link stability in high-density scenarios [130].

In addition to conceptual comparisons, the use of measurable evaluation is needed to find out the performance of legacy and emerging IoT standards in the performance dimensions. Table 3 compares NB-IoT, LTE-M, 5G RedCap, mmWave, and THz-IoT regarding range, latency, power efficiency, and density of devices. The comparison identifies the trade-offs in terms of coverage and throughput as networks are made 6G-IoT.

Figure 9 shows the variation of the path-loss in the free-space between spectrum bands applicable to IoT devices. It proves that sub-GHz frequencies still are optimal in long-range and low-power Internet of Things connections whereas mmWave and THz frequencies provide ultra-high throughput at the expense of much higher attenuation. This tendency highlights the importance of high-density small-cell coverage and 6G-IoT network energy-harvesting improvement [130, 134].

Although the 275 GHz to 3THz band has unrealized worldwide functionality, it is clear that 6G networks can benefit from its potential to achieve higher data throughput. When paired with the current mmWave band, the THz band can increase the total band capacity by at least 11.11x. The 300 GHz–3THz range is within the optical band,

Table 3 Comparative analysis of IoT connectivity technologies toward 6G

Technology	Frequency band	Range	Data rate	Latency	Device density (devices/km^2)	Power consumption	Typical use cases
NB-IoT	< 1 GHz	1–10km	≤ 100 kbps	10–100 ms	~10^5	Very Low	Smart meters, environmental sensors
LTE-M (Cat-M1)	1–2 GHz	1–5 km	≤ 1 Mbps	10 ms	~10^5	Low	Asset tracking, wearables
5G RedCap/ eRedCap	Sub-6 GHz	< 2 km	~ 100 Mbps	1–5 ms	~10^6	Medium	Industrial IoT, connected vehicles
mmWave 5G/6G	28–60 GHz	< 200 m	1–10 Gbps	< 1 ms	>10^6	High	Smart factory, VR/AR IoT
THz 6G	0.1–1 THz	< 100 m	10–100 Gbps	< 0.1 ms	>10^7	Very High	Holographic IoT, digital twins

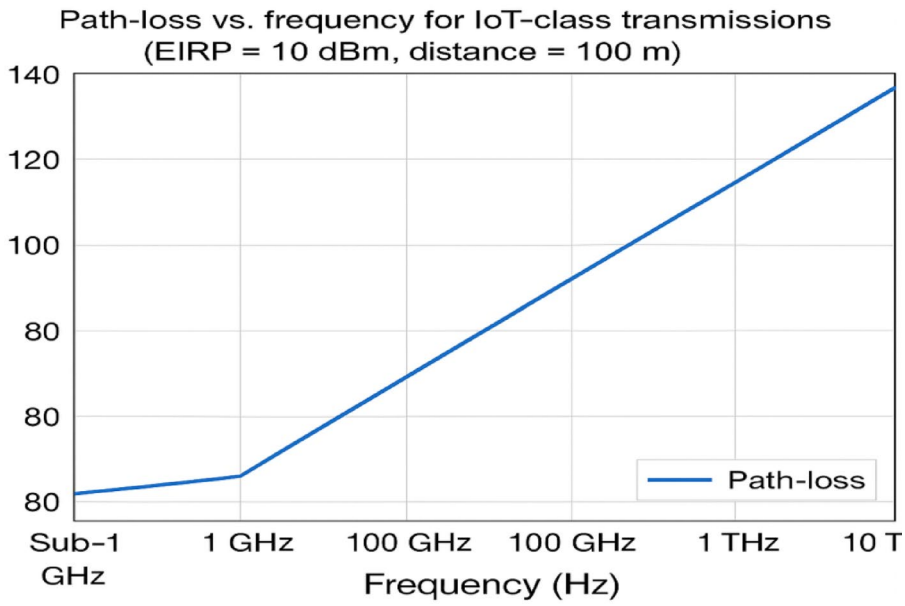

Fig. 9 Path-loss vs. frequency for IoT-class transmissions (EIRP = 10 dBm, distance = 100 m)

but because of its close closeness to the optical band boundary, its characteristics are more similar to those of the RF band. This offers prospects as well as difficulties for THz's application in 6G wireless communications. The important characteristics of THz include its wider application in reaching high data rates and its larger path loss because of its increased frequency, which makes it essential for a number of 6G services including uMUB, uHSLLC, and uHDD. Compared to the mmWave band, the THz band offers hundreds of beams and a shorter wavelength that accommodates several antennas. 6G networks can function better by utilizing orbital angular momentum (OAM) multiplexing, which can further increase overall spectrum efficiency. THz band use also makes it possible for 6G networks to quickly and effectively provide services like wireless sensing, cognition, imaging, positioning, and communication. THz is a useful tool for 6G communication systems because of its directional qualities, which when combined with pencil beam-based antennas, alleviate the problem of high atmospheric attenuation.

For IoT, THz links are best suited to short-range, high-density environments such as industrial IoT clusters. To overcome high path loss, RIS-assisted THz transceivers and energy-harvesting micro-sensors can sustain efficient, low-power connectivity.

7.13 Wireless optical communications

Visible light communication (VLC), light fidelity, optical camera communication, and optical band-based Free-Space Optical (FSO) communication are a few well-known OWC technologies that are widely used in a variety of applications (e.g., VR, underwater OWC, indoor mobile robot positioning, and V2X communication) [122, 123]. OWCs are designed for 6G communications in addition to RF-based communications, and FSO within OWC provides network-to-backhaul/fronthaul connectivity. Remote geographic areas present difficulties for optical fiber-based connectivity, and small-cell network installations of optical fiber links might not be a financially viable option. Integrating access and backhaul networks becomes essential in 6G networks because of the requirement for a high user density, and FSO fronthaul/backhaul networks are starting to show

promise as workable alternatives [25]. When it comes to data transfer, FSO-based systems function in a way that is genuinely self-directed and autonomous, much like optical fiber networks.

In 6G-IoT, visible-light and free-space optical links enable low-interference, high-capacity indoor IoT connectivity, supporting applications like precision manufacturing, healthcare monitoring, and secure data exchange.

7.14 Mimo-free-cell transmission

Two forms of intelligent surfaces that show promise for 6G technologies are large intelligent surfaces (LIS) and intelligent reflecting surfaces (IRS) [124]. LIS uses antenna arrays as surfaces in massive MIMO systems, which is different from beamforming, which needs a lot of antennas to focus signals. In contrast to beamforming, LIS avoids the detrimental effects of antenna correlations and is electromagnetically proactive in the external environment, placing less restrictions on antenna dispersion. But because LIS is active, it uses more power and is less energy-efficient.

In 6G networks, numerous frequencies and communication technologies converge to allow for smooth network transitions without the need for manual configuration. 6G adopts cell-free and non-orthogonal communications, moving away from traditional cellular and orthogonal communications. This enables autonomous network selection among the communication technologies that are available. 6G cell-free communications promise to alleviate the handover problems, delays, and data losses that occur during cell changes that plague current networks. The effective improvement of cell-free communication in 6G is facilitated by heterogeneous radio-based devices, multi-tier hybrid approaches, and multi-connectivity.

For large-scale IoT deployments, integrating blockchain with intelligent reflecting surfaces ensures secure, distributed data exchange and traceability without requiring high-power centralized servers—aligning with 6G's vision of trustworthy, energy-aware IoT.

7.15 Blockchain from security perspective

Blockchain is a tamper-proof and reversely difficult technology that operates as a decentralized database based on hash tree theory [125]. As a result, a blockchain database has characteristics like excellent data security, accessibility, and authenticity [126]. Without a central authority, blockchain can efficiently manage spectrum resources and guarantee data security in the context of 6G. For example [127], presents a privacy-preserving blockchain-based method that leverages encryption and access control technologies to protect data privacy, demonstrating flexibility for 6G-IoT. For 6G-IoT, it may also be worthwhile to investigate how blockchain, as a decentralized database, might improve accessibility protocols and ease spectral allocation in mobile cognitive radio networks [128].

8 Standards evolution toward 6G-IoT

The 6G is closely connected with the current 3GPP and ITU standardization efforts.

The 5G Advanced which is defined in 3GPP Release 18 and 19 (2023–2025) provide the following features to IoT: Reduced Capability (RedCap/eRedCap) devices, Non-Terrestrial Network (NTN) IoT integration, sidelink communication with sensors, RIS studies to improve coverage [133, 134].

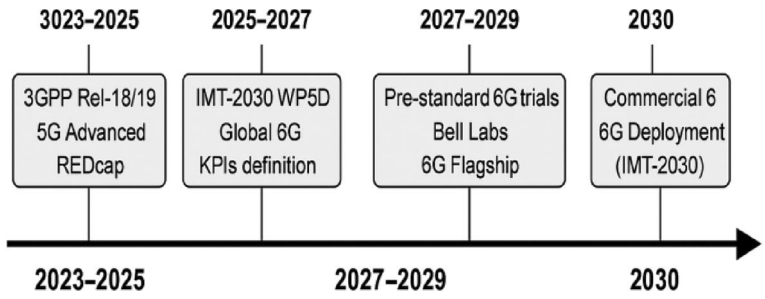

Fig. 10 Standardization roadmap linking 5G Advanced to 6G-IoT

They represent an intermediation between IMT-2020 (5G) and IMT-2030 (6G) models.

In 2023 the ITU-R Working Party 5D has introduced the IMT-2030 program (2023–2024) to specify performance and spectrum requirements of 6G systems intended for commercial deployment at around 2030 [134].

Similar initiatives include the 6G Flagship project (University of Oulu) and Bell Labs 6G work which describe the direction of AI-native networks, terahertz connections and energy conscious IoT systems [129, 135].

The standardization roadmap projected between 5G Advanced to 6G- IoT is presented in Fig. 10.

It reveals how the improvements in 3GPP Releases 18 and 19 can form the foundation of 6G, including ubiquitous intelligence, integration of the terrestrial and aerial with the satellites, and long-term IoT connections.

9 Conclusion

As wireless communication networks advance through generations, new features are added to meet changing needs and current technology are improved. The 5G communication technology is not able to handle the ever-increasing needs of wireless communication, despite its promised qualities. As a result, the development of 6G networks is essential to meeting the demands of the new communication system era. Prolonged and intensive research projects illuminate key elements of 6G networks, pointing to a bright future. A succinct synopsis of 6G networks, their development, marketing and research initiatives, supporting technologies, and the state-of-the-art in 6G communication systems is given in this overview. In addition to providing insights into the 6G concept, the story explores the different core technologies that are influencing 6G. Furthermore, focus is placed on new technologies that improve data speeds, such as innovative communication paradigms (e.g., molecular and quantum communication) and relatively contemporary spectrum technologies (e.g., THz communication and VLC). We hope that this study will encourage academics to investigate enabling technologies for 6G systems and their applications in the Internet of Things, given the widening global digital gap.

Author contributions

Aarti: Conceptualization, methodology design, data analysis, and writing – original draft. Swathi Gowroju: Software implementation, validation, visualization, and writing – review & editing. Syed Immamul Ansarullah: Resources, supervision, project administration, and final manuscript approval.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not Applicable. The manuscript does not contain any identifiable images, personal data, or information related to human participants that would require additional informed consent for publication. All authors have read and approved the final version of the manuscript and consent to its publication.

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